Arabic Startup Idea Novelty Detection Using Semantic Similarity Techniques

Ruba Aljuhani
University of Jeddah
ruba35uj@gmail.com
https://github.com/ii3ruj/Arabic-Idea-Novelty-Detection-Using-Semantic-Similarity/tree/main

Abstract

Measuring the novelty of startup ideas can be difficult, especially for Arabic content where many ideas may use different wording but still describe similar concepts. In addition, there are limited Arabic NLP systems that help users compare their startup ideas with existing ideas automatically.

This project presents an Arabic NLP system for measuring the novelty of startup ideas using semantic similarity techniques. A dataset of approximately 3756 Arabic startup ideas was collected from hackathons, competitions, Twitter/X, LinkedIn, and other public sources ,A small portion of the dataset was generated based on existing startup idea patterns to improve diversity and variation. Four NLP approaches were implemented and compared: Bag of Words (BoW), TF-IDF, Sentence-BERT (SBERT), and AraBERT.

Cosine similarity was used to calculate similarity scores between startup ideas, and novelty scores were generated based on the similarity results. The experiments showed clear differences between traditional NLP methods and transformer-based models. Traditional approaches mainly depended on word overlap, while SBERT and AraBERT were more effective in understanding semantic meaning and contextual relationships between ideas.

The results demonstrate that semantic similarity techniques can help evaluate the novelty of Arabic startup ideas more effectively and highlight the advantages of transformer-based models in Arabic NLP applications.

1 Introduction

In recent years, entrepreneurship and startup innovation have become more popular, especially in technology and AI-related fields. Many students and entrepreneurs participate in hackathons, competitions, and innovation programs to present new startup ideas and solutions. However, it is often difficult to know whether an idea is actually new or already similar to existing ideas.

Most idea evaluation is done manually, which can be subjective and time-consuming. In many cases, two startup ideas may use different words but still describe the same concept. This problem becomes more challenging in Arabic because Arabic NLP resources and semantic similarity systems are still limited compared to English.

The idea of this project came from noticing that there are very few Arabic platforms or tools that help users compare their startup ideas with existing ideas. Because of that, this project focuses on building an Arabic NLP system that measures the novelty of startup ideas using semantic similarity techniques.

In this project, a dataset of approximately 3756 Arabic startup ideas was collected from hackathons, startup competitions, Twitter/X, LinkedIn, and other public sources. Four NLP approaches were implemented and compared: Bag of Words (BoW), TF-IDF, Sentence-BERT (SBERT), and AraBERT. Cosine similarity was used to calculate similarity between ideas, and novelty scores were generated based on the similarity results.

The goal of this project is to compare traditional NLP methods with transformer-based models for measuring the novelty of Arabic startup ideas and to analyze how semantic understanding affects similarity detection in Arabic text.

2 Related Work

Text similarity and semantic similarity are important tasks in Natural Language Processing (NLP). They are used to measure how similar two texts are in meaning. Traditional approaches such as Bag of Words (BoW) and TF-IDF are widely used because they are simple and fast. However, these methods mainly depend on word overlap and frequency, which makes it difficult to capture the actual semantic meaning of the text [1].

Several studies compared traditional text representation methods such as TF-IDF and Latent Semantic Indexing (LSI) for information retrieval and text classification tasks. These studies showed that LSI can capture semantic relationships better than TF-IDF, but both methods still have limitations in understanding context and sentence meaning [1].

To improve semantic understanding, researchers introduced semantic similarity approaches that focus more on meaning instead of exact word matching. Corley and Mihalcea proposed a semantic similarity method that measures similarity between texts based on semantic relationships between words, and their results showed better performance compared to traditional lexical approaches [2].

Semantic similarity has also been explored in Arabic NLP. Nagoudi and Schwab used word embeddings for measuring semantic similarity between Arabic sentences and achieved good results compared to traditional methods [3]. In addition, previous studies discussed the challenges of Arabic semantic similarity due to Arabic morphology, writing variations, and contextual ambiguity [4].

Another Arabic NLP study combined TF-IDF and word embeddings for paraphrase identification in Arabic texts. The results showed that combining statistical and semantic representations can improve similarity detection performance [5].

Recently, transformer-based models achieved significant improvements in NLP tasks. BERT introduced deep contextual language representations using transformer architectures and achieved strong performance in many NLP benchmarks [6]. Later, Sentence-BERT (SBERT) improved sentence similarity tasks by generating sentence embeddings that can be efficiently compared using cosine similarity [7].

For Arabic NLP applications, AraBERT was introduced as a pretrained transformer model specifically designed for Arabic language understanding [8]. AraBERT achieved strong results in several Arabic NLP tasks such as text classification, sentiment analysis, and semantic similarity.

Although many studies explored semantic similarity and transformer-based models, there is still limited research focused on measuring the novelty of Arabic startup ideas. In addition, there are very few Arabic systems that automatically compare startup ideas with existing ideas. Therefore, this project compares traditional NLP methods and transformer-based approaches for evaluating the novelty of Arabic startup ideas using semantic similarity techniques.

3 Dataset

A dataset of approximately 3756 Arabic startup ideas was collected for this project. The ideas were gathered from different public sources related to entrepreneurship and innovation, including hackathons, startup competitions, Twitter/X posts, LinkedIn posts, and other publicly available startup ideas. The dataset includes ideas from different domains such as healthcare, education, artificial intelligence, smart cities, transportation, sustainability, and e-commerce.

> To increase dataset diversity, a small portion of additional startup ideas was generated based on existing startup concepts and writing patterns. These generated ideas were used only to improve dataset variety and evaluate model behavior on different idea structures.

The collected ideas varied in writing style and length. Some ideas were written in Modern Standard Arabic, while others included informal Arabic commonly used in social media and startup communities. This helped make the dataset more realistic and suitable for testing semantic similarity in real-world Arabic text.

Before applying the NLP models, several preprocessing steps were performed to clean and normalize the text data. These steps included:

- Removing punctuation and special characters
- Removing numbers and extra spaces
- Removing Arabic stopwords
- Normalizing Arabic text

After preprocessing, cosine similarity was used to compare startup ideas and calculate similarity scores.

The following thresholds were used:

- Novelty $< 0.30 \rightarrow$ Low Novelty
- $0.30 \leq$ Novelty $< 0.60 \rightarrow$ Medium Novelty
- Novelty $\geq 0.60 \rightarrow$ High Novelty

Item	Value
Number of Ideas	3756
Language	Arabic
Sources	Public Startup Sources
Domains	AI, Healthcare, Education, etc.

Table 1: Dataset Statistics

4 Methodology

This project aims to measure the novelty of Arabic startup ideas by calculating semantic similarity between a new startup idea and existing ideas in the dataset. Four different NLP approaches were implemented and compared:

- Bag of Words (BoW)
- TF-IDF
- Sentence-BERT (SBERT)
- AraBERT

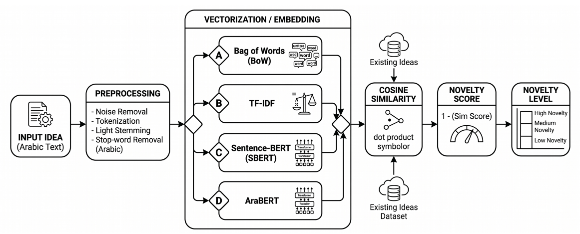

Figure 1: system pipeline.

4.1 Text Preprocessing

Before applying the models, the text data was preprocessed to improve consistency and reduce noise in the Arabic text. The preprocessing steps included:

- Removing punctuation and special characters
- Removing numbers and extra spaces
- Removing Arabic stopwords
- Normalizing Arabic letters

These preprocessing steps helped improve text quality and similarity calculations.

4.2 Bag of Words (BoW)

The first model used in this project was Bag of Words. In this approach, each startup idea was represented as a vector based on word frequency. Cosine similarity was then used to compare vectors and measure similarity between ideas.

This method is simple and fast, but it mainly depends on direct word overlap and does not understand semantic meaning.

4.3 TF-IDF

TF-IDF (Term Frequency–Inverse Document Frequency) was used as an improved traditional NLP approach. Unlike Bag of Words, TF-IDF gives more importance to informative words and reduces the effect of common words.

Each startup idea was converted into a TF-IDF vector, and cosine similarity was calculated between ideas to measure similarity.

4.4 Sentence-BERT (SBERT)

Sentence-BERT (SBERT) was used to generate semantic sentence embeddings for startup ideas. SBERT can better understand contextual meaning compared to traditional vectorization methods.

The generated embeddings were compared using cosine similarity to identify semantically related startup ideas, even when different wording was used.

4.5 AraBERT

AraBERT was used as an Arabic transformer-based model specialized for Arabic language understanding. The model generated contextual embeddings for startup ideas and improved semantic similarity detection in Arabic text.

Compared to traditional approaches, AraBERT provided better understanding of Arabic context and semantic relationships between ideas.

4.6 Similarity and Novelty Calculation

Cosine similarity was used in all models to measure similarity between startup ideas. After calculating similarity, the novelty score was computed using the following equation: The novelty score was calculated using the following equation:

$$NoveltyScore = 1 - CosineSimilarity$$

Based on the novelty score, ideas were classified into three novelty levels:

- Low Novelty
- Medium Novelty
- High Novelty

based on predefined novelty thresholds.

5 Experiments and Evaluation

Several experiments were conducted to compare the performance of traditional NLP methods and transformer-based models in measuring the novelty of Arabic startup ideas. The experiments were implemented using Python in Google Colab.

The evaluation focused on comparing similarity scores and novelty levels generated by each model. Different startup ideas from multiple domains were tested to analyze how each model understands semantic similarity in Arabic text.

Component	Details
Programming Language	Python
Environment	Google Colab
Libraries	Scikit-learn, Transformers
Dataset Size	3756 Ideas

Table 2: Experimental Environment

5.1 Experimental Example

5.1.1 Example 1

The following startup idea was tested using the implemented NLP models:

منصة استشارات فورية وحجز ذكي للمواعيد الرياضية

Model	Similarity	Novelty	Level
BoW	0.50	0.50	Medium
TF-IDF	0.59	0.40	Medium
SBERT	0.58	0.41	Medium
AraBERT	0.59	0.40	Medium

Table 3: Results for sports consultation platform idea

Analysis

All models produced medium novelty scores because the idea shares common concepts related to booking systems and digital consultation platforms. The results indicate that similar startup ideas already exist in different domains and applications.

5.1.2 Example 2

The following startup idea was tested:

نظام يستخدم الرؤية الحاسوبية لتحليل إشارات سائقي الشاحنات لمنع الحوادث

Model	Similarity	Novelty	Level
BoW	0.14	0.85	High
TF-IDF	0.16	0.83	High
SBERT	0.75	0.24	Low
AraBERT	0.74	0.25	Low

Table 4: Results for computer vision safety system idea

Analysis

Traditional NLP methods produced high novelty scores because the idea contained specialized terms and low direct word overlap with existing ideas. However, SBERT and AraBERT detected semantic similarity with existing AI and computer vision concepts, resulting in lower novelty levels.

5.1.3 Example 3

The following startup idea was tested:

منقي للتربة يعمل على ضخ مواد طبيعية لتحويلها إلى تربة عضوية

Model	Similarity	Novelty	Level
BoW	0.15	0.84	High
TF-IDF	0.17	0.82	High
SBERT	0.62	0.37	Medium
AraBERT	0.73	0.26	Low

Table 5: Results for soil purification idea

Analysis

The idea achieved high novelty scores in traditional NLP models due to low lexical overlap with existing ideas. In contrast, transformer-based models identified semantic relationships with sustainability and environmental technology concepts, which reduced the novelty level.

5.1.4 Example 4

The following startup idea was tested:

روبوت يتنكر عى هيئة البشر

Model	Similarity	Novelty	Level
BoW	0.11	0.88	High
TF-IDF	0.13	0.86	High
SBERT	0.58	0.41	Medium
AraBERT	0.57	0.42	Medium

Table 6: Results for robot identity imitation idea

Analysis

Traditional NLP models produced high novelty scores because the idea used uncommon vocabulary with low direct lexical overlap. However, SBERT and AraBERT detected semantic similarity with existing AI and robotics concepts, resulting in medium novelty levels.

5.1.5 General Discussion

The experiments showed noticeable differences between traditional NLP approaches and transformer-based models. In several cases, traditional methods such as Bag of Words and TF-IDF produced high novelty scores because they mainly depended on direct word overlap. In contrast, SBERT and AraBERT were more effective in detecting semantic relationships between startup ideas, even when different wording was used.

The results also showed that startup ideas may receive different novelty levels depending on the NLP approach. Traditional methods mainly focused on lexical similarity, while transformer-based models captured contextual and semantic meaning more effectively.

Overall, the findings demonstrate the importance of semantic understanding in evaluating the novelty of Arabic startup ideas and highlight the advantages of transformer-based models in Arabic NLP applications.

5.2 Runtime Comparison

The runtime of the models was compared during testing. Traditional NLP approaches such as Bag of Words and TF-IDF produced results faster because they rely on simple vectorization techniques. In contrast, transformer-based models such as SBERT and AraBERT required more processing time due to embedding generation and contextual representation.

Model	Runtime
BoW	Fast
TF-IDF	Fast
SBERT	Moderate
AraBERT	Moderate

Table 7: Runtime comparison between the implemented models

5.3 Discussion

The experimental results confirmed that transformer-based models achieved stronger semantic understanding compared to traditional lexical-based approaches. While Bag of Words and TF-IDF relied mainly on direct keyword overlap, SBERT and AraBERT were more effective in capturing contextual relationships between startup ideas.

The experiments further demonstrated that semantic similarity techniques can improve novelty evaluation for Arabic startup ideas, especially when ideas are described using different wording but share similar concepts.

Overall, AraBERT and SBERT provided more reliable semantic similarity results, while traditional NLP methods remained faster and computationally simpler.

6 Conclusion

This project presented an Arabic NLP system for measuring the novelty of startup ideas using semantic similarity techniques. A dataset of approximately 3756 Arabic startup ideas was collected from hackathons, competitions, Twitter/X, LinkedIn, and existing startup idea sources. A small portion of additional ideas was generated to improve dataset diversity and variation.

Four NLP approaches were implemented and compared: Bag of Words, TF-IDF, Sentence-BERT (SBERT), and AraBERT. Cosine similarity was used to measure similarity between startup ideas, and novelty scores were calculated based on the similarity results.

The experiments showed clear differences between traditional NLP methods and transformer-based models. Traditional approaches such as BoW and TF-IDF mainly relied on direct word overlap, while SBERT and AraBERT were more effective in understanding semantic meaning and contextual relationships between ideas.

The results demonstrate that transformer-based models provide better semantic similarity performance for Arabic startup ideas and can improve novelty evaluation compared to traditional NLP methods.

One limitation of this project is that some startup ideas were general and shared similar concepts, which occasionally produced high similarity scores across different domains. In future work, this project can be extended by collecting larger Arabic datasets, improving preprocessing techniques, and developing a web-based platform that automatically compares startup ideas and estimates their novelty levels.

7 References

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